

QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE3003

Course Title: Computer Networks

Set number: 5

Date of Exam: 29/03/2023 (AN)

Duration: 90 Minutes

Total Marks: 50 (C2)

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. a) Suppose we want to transmit the message “1001100100001” and employ CRC algorithm with $G(x) = x^3 + 1$ to make sure it reaches receiver without any errors. Find the valid codeword.

(5M)

b) Alice wants to send a “010010010001100101101011001001” message to Bob. However another person named Joe intercepts the message and modifies some bit(s). When Bob receives the message, he want to verify the integrity of the message so he applies two dimensional parity check. For implementing parity check, he assumes even parity. Show the mechanism of the approach in 2D parity check.

(5M)

Q2. a) A transmission channel has a bandwidth of 8.2 kbps and one way propagation delay of 47 ms. The system uses stop and wait protocol for transmitting messages. Assuming there is negligible delay in receipt of acknowledgements for every successful transfer, the system intends to achieve an efficiency of at least 60%. What should be the minimal frame size for achieving the given efficiency?

(5 M)

b) Consider a 256×10^3 bits/second communication channel with propagation delay of 150 ms. Selective repeat protocol is used on this link to send the data with frame size of 1 KB. Neglect the time of Acknowledgement packet. What is the minimum number of packets to achieve link utilization of 80% ?

(5 M)

Q3. a) Suppose there are “x” slots of slotted ALOHA. Each node attempts to transmit the data with probability of “p” in each time slot. For what value of p, efficiency of slotted ALOHA can be maximum?

(5 M)

b) A frame of 3000 bytes travel through six switches along the path. Each link has a bandwidth of 200 mbps, length of 230 m and propagation speed of 3.5×10^8 m/s. Assuming processing delay of 5 ms at each switch, what is the approximate total time for transmitting the packet ?

(5 M)

Q4. (a) Explain the functionality and bit size of the following fields in IPV4 header. (2 x 3 M = 6 M)

- i. Fragment offset
- ii. Options and padding
- iii. Type of service

(b) For an address 205.16.37.39/28, find first address and number of addresses.

(4 M)

Q.5. a) A packet bearing a destination address 144.16.68.117 arrives at the router. What are the possible interfaces to which the packet will be forwarded ? **(8 M)**

Destination	Subnet Mask	Interface
147.16.0.0	255.255.255.0	ETH0
147.16.64.0	255.255.255.255	ETH1
147.16.68.0	255.255.255.255	ETH2
147.16.68.64	255.255.255.224	ETH3

b) In IPV4 addressing format, how many networks are allowed under class C addressing ? How many host IDs will be formed ? **(2 M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	1	1, 2	1	1	10
Q2	3	2	2	1	1	10
Q3	3	2	3, 4	1	2	10
Q4	4	4	4,5	3	2	10
Q5	4	4	4	4	2	10

QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE3003

Course Title: Computer Networks

Set number: 7

Date of Exam: 27/03/2023

Duration: 90 Min

Total Marks: 50 (FN)(A1)

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. a) Consider the Cyclic Redundancy Check (CRC) algorithm with the 4-bit generator polynomial (G) x^3+1 and the data is $x^7+ x^4+ x^3+ x^2+x+1$. Calculate the redundant bits that sender will add to the data while sending to destination. **(5M)**

b) The checksum value is 1011 for the following data bits where there are 4 sections and 4 bits in each section. Find the value of x. **(5M)**

1001 1100 1010 001x

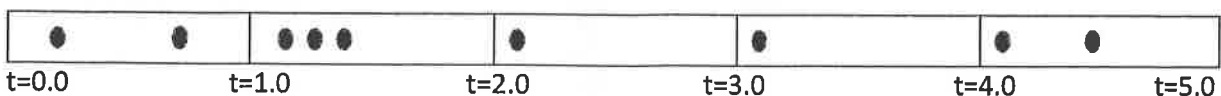
Q2. a) A 3000 km long trunk is used to transmit using Go-Back-N protocol. The propagation speed is 6 microseconds/km and trunk data rate is 1.544 Mbps. Ignore the time taken to receive the bits in the acknowledgement. Frame size is 64 Bytes. If Go-Back-N protocol is used, in order to achieve of 100 %, what is the maximum window size at the sender's side? **(4 M)**

b) Host A wants to send 8 frames to Host B. The host agreed to go with Go-Back-3. How many number of frames are transmitted by Host A if every 3rd frame that is transmitted by A is corrupted or lost? Compare the number of transmissions of SR ARQ with Go-Back-3 ARQ. Consider the window size for SR ARQ is 3. **(6 M)**

Q.3. a) Consider the figure below, which shows the arrival of 9 messages at different 9 multiple nodes as $t = \langle 0.3, 0.7, 1.2, 1.3, 1.4, 2.1, 3.2, 4.1, 4.5 \rangle$ respectively. It means that these 9 nodes have a ready message to send at this respective time. Assume each transmission of message requires exactly one time unit. Suppose all nodes are implementing the slotted Aloha protocol then

i) Indicate the time at which transmission of each message begins.

ii) Which messages will transmit successfully? Write down the number. **(5 M)**



b) Suppose nodes A and B are on same 10 Mbps Ethernet segment using CSMA/CD and minimum size of the packet is 225 bits. Suppose A and B send frames at $t=0$, the frames collide, then at what time, they finish transmitting a jam signal? Assume a 48 bit jam signal. Note that jam signal is sent/release by nodes immediately as soon as they received the collision signal for collision. Also, note that jam

signal alerts the other stations not to transmit their data immediately after the collision. Also, calculate at what time collision occurs. (5 M)

Q.4. (a) Draw the diagram of IPv4 datagram format. Explain the Header checksum, Fragmentation offset and Time-to-Live terms (7 M)

(b) Identify the Class A and Class B Host address from following set of addresses. 128.4.5.6, 117.4.5.1, 117.0.0.0, 117.8.0.0, 230.0.0.0, 130.4.5.6, 30.4.5.6 (3 M)

Q.5. A class B network address 130.50.0.0 is submitted as follows. The last 10 bits of the host id are allotted for host number and the remaining 6 bits are reserved for subnet number. How many subnets and number of hosts in each subnet are possible with the above addressing scheme? Also, Find out the starting address of 1st, 3rd and 5th subnets ? (10 M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	1	1, 2	1	1	10
Q2	3	2	2	1	1	10
Q3	3	2	3, 4	1	2	10
Q4	4	4	4,5	3	2	10
Q5	4	4	4	4	2	10

QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE3003

Course Title: Computer Networks

Set number: 8

Date of Exam: 28/03/2023 (An)

Duration: 90 Min

Total Marks: 50 (B₂)

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. a) Consider the 12-bit Hamming codeword as **D8 D7 D6 D5 P8 D4 D3 D2 P4 D1 P2 P1** where there is a 8 bit data bits and 4 parity bits which are given in the following tables. Find out the value of x and y. Assume that even parity is considered for finding out parity bits. **(5 M)**

Parity Bits

Data Bits

P8	P4	P2	P1	D8	D7	D6	D5	D4	D3	D2	D1
y	0	1	0	1	1	0	x	0	1	0	1

b) Consider 2D parity check with the following data of 5 rows and 16 columns. Find the out the even parity bits for it. Suppose two errors occurs at (R3, C4) and (R3,C6). It means bits are changed at these locations. Show that how these two errors can be detected using 2D Parity Check. **(5 M)**

1	0	0	1	1	0	1	1	0	1	0	1	0	0	0	0
0	1	0	0	1	0	0	0	1	1	0	1	0	1	1	1
1	1	1	1	0	0	0	0	1	1	1	0	1	1	1	1
1	1	1	1	1	0	0	1	1	1	1	1	1	1	1	1
1	0	0	1	1	1	1	0	1	0	0	1	1	0	0	1

Q2. a) Consider two hosts A and B located at two different cities in India. The channel is operating at 5000 bps and the round trip time between the two end systems is 30 ms. What would be the minimum frame size for stop and wait ARQ flow control protocol to get 60 % link utilization efficiency? **(3 M)**

b) Station A needs to send a message consisting of 12 packets to Station B using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 4th packet that A transmits gets lost (but no acks from B ever get lost), then what is the number of packets that A will transmit for sending the message to B? **(7 M)**

Q.3. a) What are the three strategies of CSMA, explain it in brief. Identify the strategy that has high chance of collision. Also identify the strategy that less chance of collision among all these three strategies. **(5 M)**

b) A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The minimum packet size that can be used on this network is 200 bits. Find out speed of propagation in m/sec. Consider there are 2 stations on this 2 km LAN. Suppose A and B send frames at $t=0$. Does the frames collide and if collided, then calculate the time at which frames collided and at what time stations A and B received the collided signal. **(5 M)**

Q.4. (a) What is Classful and Classless addressing? Describe all the classes of Classful addressing in brief. Why Classless addressing is needed? **(7 M)**

(b) Which of the following IP addresses are valid to be assigned to the host **(3 M)**

192.248.16.255, 25.5.25.55, 150.7.0.0, 23.56.9.43, 192.168.4.5

Q.5. a) A router uses the following routing table.

Destination	Mask	Interface
144.16.0.0	255.255.0.0	Eth 0
144.16.64.0	255.255.254.0	Eth 1
144.16.68.0	255.255.255.0	Eth 2
144.16.68.64	255.255.255.224	Eth 3

A packet bearing a destination address 144.16.68.117 arrives at the router. On which interface it will be forwarded. **(5 M)**

b) Consider following IP address and subnet mask.

IP address=172.60.50.2, Subnet mask=255.255.224.0

Find the range of assignable IP address on the subnet in which host belongs. **(5 M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	1	1, 2	1	1	10
Q2	3	2	2	1	1	10
Q3	3	2	3, 4	1	2	10
Q4	4	4	4,5	3	2	10
Q5	4	4	4	4	2	10